STAFF REVIEW · H446 221 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. f=7.

2. It lets us reconstruct the path by following predecessors backwards.

3. 7.

4. No. It overestimates the true cost.

I Do — source worked steps

1. Predict the choice — V has g1,h3,f4; W has g5,h1,f6. Expand V.

2. Review the choice — V improves W to g3 and f4; expand W.

3. Review the choice — W improves the goal G to g5,h0,f5.

4. Review the choice — The path is U→V→W→G with total cost 5; the heuristics do not exceed the actual remaining costs.

We Do — guided CHECK

1. Review the choice
O reaches G at g7,f7. G is selected before N, so the early-goal algorithm returns cost 7.
But M→N→O→G costs 2+2+2=6; the bad h(N)=7 overestimates N's true remaining cost 4.
Using an admissible estimate such as h(N)=3 allows the cheaper path to be found.

You Do — source mark schemes

1. [4] h(A)=4 is admissible [1] because it does not exceed 6 [1]; h(A)=8 is inadmissible [1] because it overestimates 6 [1].

2. [6] Route S→A→B→G [2], cost 6 [1]; from A the true cheapest remaining cost is 4 [1], so h(A)=9 overestimates [1] and can delay expanding A while a more expensive goal route appears [1].

3. [6] Dijkstra selects lowest tentative g [1] and does not need h [1]; A* selects lowest f=g+h [2]; Dijkstra needs non-negative edges [1]; A* requires suitable admissible heuristic (and appropriate closed-set handling, usually consistency) to guarantee optimality [1].

Exit Ticket — source answer

1. True. An inadmissible estimate can make a less costly path appear too unattractive to explore before stopping.